Computers in Biology and Medicine 140 (2022) 105105

Contents lists available at ScienceDirect

Computers in Biology and Medicine

journal homepage: www.elsevier.com/locate/compbiomed

Fully automated segmentation of lumbar bone marrow in sagittal, high-resolution T1-weighted magnetic resonance images using 2D U-NET

Eo-Jin Hwang [a], Sanghee Kim [b], Joon-Yong Jung [a,*]

[a] Department of Radiology, Seoul St. Mary's Hospital, College of Medicine, The Catholic University of Korea, Seoul, Republic of Korea
[b] Department of Radiology, Yeouido St. Mary's Hospital, College of Medicine, The Catholic University of Korea, Seoul, Republic of Korea

ARTICLE INFO

Keywords:
Magnetic resonance imaging
Lumbar bone marrow
Hematologic diseases
Deep learning segmentation and image segmentation model

ABSTRACT

Background: We investigated a 2-dimensional (2D) U-Net model to delineate lumbar bone marrow (BM) using a high resolution T1-weighted magnetic resonance imaging.
Method: Healthy controls (n = 44, 836 images) and patients with hematologic diseases (n = 56, 1064 images) received MRI of the lumbar spines. Lumbar BM on each image was manually delineated by an experienced radiologist as a ground-truth. The 2D U-Net models were trained using a healthy lumbar BM only, diseased BM only, and using healthy and diseased BM combined, respectively. The models were validated using healthy and diseased subjects, separately. A repeated-measures analysis of variance was performed to compare segmentation accuracies with 2 validation cohorts among U-Net trained with healthy subjects (UNET_HC), U-Net trained with diseased subjects (UNET_HD), U-Net trained with all subjects including both healthy and diseased subjects (UNET_HCHD), and 3-dimensional Grow-Cut algorithm (3DGC).
Results: When validated with the healthy subjects, UNET_HC, UNET_HD, UNET_HCHD and 3DGC achieved the mean and standard deviation of the Dice Similarity Coefficient (DSC) of 0.9415 ± 0.07056, 0.9583 ± 0.05146, 0.9602 ± 0.0486 and 0.9139 ± 0.2039, respectively. When validated with the diseased subjects, DSCs of UNET_HC, UNET_HD, UNET_HCHD and 3DGC were 0.8303 ± 0.1073, 0.9502 ± 0.0217, 0.9502 ± 0.0217 and 0.8886 ± 0.2179, respectively. The U-Net models segmented BM better than the semi-automatic 3DGC ($P < 0.0001$), and UNET_HD produced better results than UNET_HC ($P < 0.0001$).
Conclusions: We successfully constructed a fully automatic lumbar BM segmentation model for a high-resolution T1-weighted MRI using U-Net, which outperformed most of the previously reported approaches and the existing semi-automatic algorithm.

1. Introduction

Bone marrow (BM) is the primary organ producing blood cells. BM is found in the spine, pelvic bone and proximal femurs and consists of heterogeneous components including fat, water and various type of blood cells. Due to its proliferating capacity and tight connectivity to systemic circulation, both primary hematologic malignancy and secondary metastatic diseases are frequently involved in BM [1]. When the hematologic malignancy develops in BM, the fat component is replaced by cancer cells, and magnetic resonance imaging (MRI) is often chosen for bone marrow evaluation because it is sensitive to visualize fat component. Particularly, the T1-weighted image in which fat appears as high signal intensity while the other components exhibit low signal intensity is the most important sequence to assess BM diseases [2]. Although many researchers have developed a number of machine learning algorithms to support BM assessment [3–6], it is often challenging to recognize subtle changes of signal intensity in BM.

Robust localization and segmentation of lumbar spine are crucial steps in analyzing BM signals and diagnosing BM diseases accurately. Attempts have been made to delineate lumbar BM for various purposes, including quantitative calculation of pure red marrow metabolism using 2-deoxy-2-[F-18]fluoro-D-glucose (FDG) positron emission tomography (PET) [7], computer-assisted surgery of the lumbar spine [8], detection of osteoporosis [9], assessment of vertebral fractures [10], characterization of the marrow compartments [11] and identification of BM alteration in the context of lower back pain and neuromuscular diseases using computer tomography (CT) or magnetic resonance imaging (MRI)

* Corresponding author. Department of Radiology, Seoul St. Mary's Hospital, College of Medicine, The Catholic University of Korea, 222 Banpo-daero, Seocho-gu, Seoul, 06591, Republic of Korea.
E-mail address: jy.jung@songeui.ac.kr (J.-Y. Jung).

https://doi.org/10.1016/j.compbiomed.2021.105105
Received 23 September 2021; Received in revised form 25 November 2021; Accepted 29 November 2021
Available online 1 December 2021
0010-4825/© 2021 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

List of abbreviations

MRI	magnetic resonance imaging
BM	bone marrow
PACS	picture archiving communication system
DSC	dice similarity coefficient
JI	Jaccard index
HSD	Hausdorff surface distance
MGUS	Monoclonal gammopathy of unknown significance
MDS	Myelodysplastic syndrome
TR	repetition time
TE	echo time
Hz	hertz
2DT1TSE	2-dimensional T1 Turbo Spin Echo
GPU	graphical processing unit
GB	gigabytes
RAM	random access memory
ADAM	adaptive moment estimator
UNET_HC[1]	The first U-Net model trained with healthy controls
UNET_HD[2]	The second U-Net model trained with diseased subjects
UNET_HCHD[3]	The third U-Net model trained with all subjects
3DGC[4]	The fourth segmentation model with a 3-dimensional Grow-Cut algorithm
rANOVA	repeated-measures analysis of variance

[12]. A number of segmentation methods and models, either semi-automatic [13–18] or fully automated, have been proposed for multiple imaging modalities including X-ray images [19], CT [20–23] and MRI [24,25].

However, previous studies did not consider how the disease status of BM affected the segmentation results. The borders between BM and the surrounding tissues become indistinct when the BM signal is heterogeneously decreased with hematologic malignancy. Hematologic diseases may also lead to various spinal problems including vertebral fractures, which may appear as endplate alternations, wedge deformities or vertebral collapse [26]. Since the altered BM signal intensity and shape clearly distinguish itself from the normal, rectangular-shaped BM, it is important to construct a segmentation model that accounts for the variability of the individual BM status. We hypothesized that segmentation models trained with normal BM would produce different segmentation results from those trained with the diseased BM with vertebral fractures.

We developed a fully automatic segmentation model of lumbar BM on sagittal, high resolution T1-weighted MRI using U-Net, a convolutional neural network used for image segmentation with a symmetrical U-shape with a left-hand contracting path to capture context of the input image followed by an expansive right-hand path to precisely localize the contextual information from the contracting path [27]. Therefore, the purpose of our study was to construct a fully automatic segmentation model for lumbar BM using a high resolution T1-weighted MRI. To determine the most suitable model for lumbar BM delineation, we trained several U-Net models with different subject groups and observed the effect of training cohorts on segmentation results. We also compared model performances with a traditional semi-automatic segmentation method to demonstrate overall improvement in segmentation process.

2. Materials and methods

This study is a cross-sectional, retrospective, and observational study performed in a single institution (Seoul St. Mary's hospital, Seoul, South Korea). This study was approved by the institutional review board, and informed consent was waived.

2.1. Inclusion and exclusion criteria for subjects

We separately collected healthy and diseased subjects from Picture Archiving Communication System (PACS) among the patients who received spine MRI between March 2010 and June 2019. For healthy subjects, inclusion criteria were 1) females aged more than 40 years old and males more than 30 years old, 2) absence of BM pathology, which was verified based on the laboratory results and clinical follow-up. Inclusion criteria for age was based on the fact that young adults and middle-aged women tend to have relatively higher cellularity in BM, which appears similar to the diseased BM. For a diseased subject group, inclusion criteria were 1) patients who visited the hematology department of our hospital, 2) patients with confirmative diagnosis of hematologic diseases based on the clinicopathological criteria, and 3) treatment-naïve patients.

Exclusion criteria were 1) severe scoliosis or congenital deformity, 2) combined inflammatory or infectious disease, 3) previous spinal surgery, and 4) severe artifacts degrading signal intensity of BM. These were applied to both groups in common.

2.2. Characteristics of finally included subjects

We randomly selected 44 healthy subjects and 56 diseased subjects who met the above criteria. The types of hematologic diseases of 56 disease subjects were leukemia (n = 30), multiple myeloma (n = 16), lymphoma (n = 5), monoclonal gammopathy of unknown significance (MGUS) (n = 3) and myelodysplastic syndrome (MDS) (n = 2). The demographic data between healthy and diseased groups were compared (Table 1). Of the 100 subjects, 22 healthy and 28 diseased subjects were randomly selected for training while the remaining 22 healthy and 28 diseased subjects were used for validation.

2.3. Magnetic resonance image acquisition

MR images were acquired using multiple MRI scanners in our institution. The sagittal, high resolution T1-weighted images included in this study were heterogeneous in terms of manufacturers, model names, magnetic fields and scanning parameters. Table 2 summarizes the sequence protocols and parameters used to acquire images.

2.4. Semi-automated lumbar bone marrow segmentation using a Grow-Cut algorithm

For a semi-automated lumbar bone marrow segmentation, we used a 3D Grow-Cut algorithm, which is a semi-automatic method to segment the area of interest from multiple slices of an image [28]. This algorithm performs a multi-label segmentation of N-dimensional images; once a user selects a few number of pixels inside the object of interest as an

Table 1
Subject characteristics of the healthy controls and patients with hematologic diseases included in the study.

	Healthy (n = 44)	Diseased (n = 56)	P-value
Age			
Mean ± standard deviation (years)	66.48 ± 9.766	62.30 ± 10.95	0.0800
Range	44–81	36–83	
Gender			
Male/Female (% Female)	19/25 (56.81)	29/27 (48.21)	0.111
Body mass index ± standard deviation (kg/m^2)	24.4 ± 3.5	23.4 ± 3.7	0.188

Table 2
Summary of MR image acquisition parameters from multiple MR vendors.

	MR1	MR2	MR3	MR4	MR5	MR6	MR7
Manufacturer	Philips	Siemens	Siemens	Siemens	Philips	Philips	GE
Model name	Ingenia	Verio	Verio	Verio	Achieva	Avanto	Signa HDxt
Scanning sequence	2DT1TSE	2DT1TSE	2DT1TSE	2DT1TSE	2DT1TSE	2DT1TSE	2DT1TSE
Magnetic field (Tesla)	3	3	3	3	1.5	1.5	1.5
TR (msec)	700	700	797	750	531.1	450	585.3
TE (msec)	10	10	10	10	20	10	21.36
Flip angle (°)	90	128	128	128	90	150	90
Echo train length	5	3	3	3	5	3	3
Resolution (mm^2)	0.59x0.59	0.55x0.55	0.625x0.625	0.55x0.55	0.55x0.55	0.67x0.67	0.51x0.51
Slice thickness (mm)	3	3	3	3	3	3	3
Gap (mm)	3	3	3	3	3	3.3	3
Matrix size	512x512	512x512	448x448	512x512	512x512	448x448	512x512
Number of slices	21	19	19	19	19	19	19
Bandwidth (Hz/pixel)	291	250	250	250	154.8	172	108.5
Number of averages	1	1	1	1	3	2	0.5

object label and those outside the object as a background label, the rest of the image is segmented automatically by Cellular Automaton [29]. Fig. 1 illustrates the segmentation process of a 3D Grow-Cut algorithm.

For each subject, lumbar BM from the sagittal T1-weighted image was manually segmented by a board-certified musculoskeletal radiologist with 12 years of experience. Fig. 2 illustrates the sagittal T1-weighted images of the healthy and diseased lumbar spines and the segmented BM regions by an experienced radiologist and by the 3D Grow-Cut algorithm, respectively.

2.5. Fully automated lumbar bone marrow segmentation using 2D U-Net

The lumbar spinal T1-weighted MRI of each subject was resampled to have identical image sizes across all subjects, and pixel values were normalized between 0 and 1. The mask images, which contain binary images with manually segmented lumbar BM as 1 and the rest of the image as 0, were aligned with the original T1-weighted MRI of the same subject. We trained the segmentation model with different types of data set in order to examine the effects of individual lumbar BM shapes on segmentation accuracies. We independently trained U-Net models with BM of the healthy subjects only (n = 22, number of slices = 418), those of the diseased subjects only (n = 28, number of slices = 532) and those of the healthy and diseased subjects combined (n = 50, number of slices = 950). The data within the training set were augmented to improve a network performance by artificially producing a greater number of data than the given number of data. In this study, we applied simple transformations such as flipping, rotation, shift and zoom to cause image displacement. Shear operation was also applied to slightly distort the global shape of lumbar BM. The batch size was 32, which was the number of augmented images produced for each iteration; a total of 100 epochs were performed for training each model.

The 2D implementation of U-Net was designed to receive augmented images in sizes of 528 × 528 pixels with 19 slices per subject and were trained on Radeon Pro Vega 64X graphics processing unit (GPU) with 16 gigabytes (GB) of random access memory (RAM), using Adaptive Moment Estimation (ADAM) as an optimizer and a learning rate of 1×10^{-2} during 100 epochs. Fig. 2 summarizes the 2D U-Net architecture used for lumbar BM segmentation. The U-Net architecture consisted of a contracting path with 5 convolutional blocks and an expanding path with the same number of blocks. Each block had two convolutional layers with a filer size of 3 × 3, stride of 1 × 1 and a rectifier activation. For the contracting path, max pooling with stride 2 × 2 was applied to the end of every block except for the last block until the size of the feature maps decreased from the original size of 528 × 528 pixels to 33 × 33, while the number of feature maps increased from 1 to 512. In the expanding path, every block started with a deconvolutional layer with a filter size of 3 × 3 and stride of 2 × 2, which doubled the sizes of feature maps while decreasing the number of feature maps by two until the size of feature maps increased from 33 × 33 to 528 × 528. The number of feature maps were reduced from 512 to 1 during the expanding path. Finally, each lumbar BM on a validation set was segmented using the trained model. Fig. 3 describes the overall architecture of the 2D U-Net model.

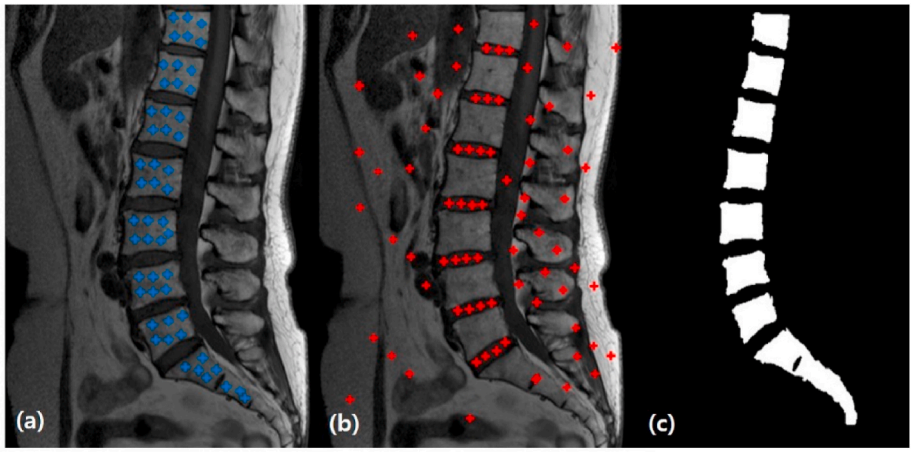


Fig. 1. Segmentation process of a 3D Grow-Cut algorithm. (a) A few pixels inside the object-of-interest are selected as denoted by blue crosses, (b) and pixels outside the object-of-interest are selected as denoted by red crosses. (c) Only the regions of interest are segmented using a 3D Grow-Cut algorithm.



Fig. 2. Sagittal, high-resolution T1-weighted MRI of lumbar spines and the segmented lumbar BM. (a) A single slice image of a healthy subject, (b) manually segmented bone marrows by an experienced radiologist (ground-truth), (c) semi-automatically segmented bone marrows of the same slice using the 3D Grow-Cut algorithm, (e) a single slice image of a diseased subject, (f) manually segmented bone marrows by an experienced radiologist (ground-truth), and (g) semi-automatically segmented bone marrows of the same slice using the 3D Grow-Cut algorithm.

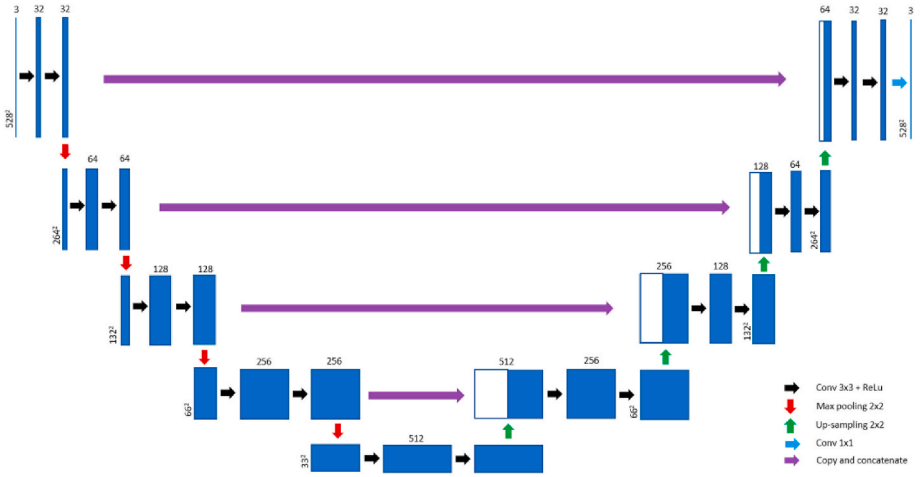


Fig. 3. Overall architecture of the 2-Dimensional U-Net model for fully automatic lumbar BM segmentation.

2.6. Performance evaluation

The performances of U-Net were evaluated qualitatively and quantitatively using the healthy control subjects (n = 22) and the diseased subject group (n = 28), separately. For quantitative assessment, dice similarity coefficient (DSC), the Jaccard Index (JI) and Hausdorff surface distance (HSD) were calculated from each slice and from the individual subjects, respectively.

$$DSC(X,Y) = \frac{2|X \cap Y|}{|X| + |Y|} \quad [1]$$

$$JI(X,Y) = \frac{|X \cap Y|}{|X \cup Y|} = \frac{|X \cap Y|}{|X| + |Y| - |X \cap Y|} \quad [2]$$

$$HSD(X,Y) = \max_{x \in X}\left\{\min_{y \in Y}\{D(x,y)\}\right\} \quad [3]$$

The DSC (Equation (1)) and JI (Equation (2)) are measures for

evaluating overlapping volumes between the segmentation results and ground truth; HSD measures the discrepancy between the segmented surface and true surface, which is defined as the maximum distance of a data set to the nearest point in the other data set where x and y are points of sets X and Y, respectively, and D (x, y) is a distance between these points (Equation (3)). For each slice, DSC, JI and HSD were calculated to compare the in-plane lumbar BM segmentation with the ground-truth performed by a radiologist.

2.7. Statistical analyses

The independent samples *t*-test was used to compare mean ages of the healthy and diseased groups, and the Mann-Whitney *U* test was performed to compare gender differences between the healthy and diseased groups.

A repeated-measures analysis of variance (rANOVA) was performed to evaluate statistical differences among the mean segmentation accuracies of the healthy and diseased BM when repeatedly segmented using the four different models: UNET_HC[1], UNET_HD[2], UNET_HCHD[3], and 3DGC[4]. For our purpose of the study, the within-subject factors were the three evaluation metrics (DSC, JI, HSD), and the between-subject factors were the healthy and diseased groups. The tests of between-subject effects and within-subject effects were analyzed using the F-value, which is a ratio of a between-subject factor variance and the within-subject factor variance. For the test of the within-subject effects, Bonferroni correction was performed for a post-hoc pairwise comparison. Except for the pairwise comparison, P value of 0.05 or less was considered as a significant difference. All statistical analyses were performed using MedCalc for Windows, version 19.4 (MedCalc Software, Ostend, Belgium).

3. Results

3.1. Subjects

The independent samples *t*-test showed that there was no statistically significant difference between mean ages and body mass indices of the healthy and diseased groups (P = 0.0800 and P = 0.188 respectively). No statistically significant difference was observed between gender distribution of the two groups (P = 0.111) (Table 1).

3.2. BM segmentation results of the healthy group per individual subject

Table 3 summarizes evaluation metrics calculated from individual subjects when validated with the healthy and diseased subjects, separately. When validated with the 22 healthy subjects, the means and standard deviations of DSC, JI and HSD for UNET_HC[1] were 0.9415 ± 0.01139, 0.8976 ± 0.01762 and 1.605 ± 1.883, respectively; those for UNET_HD[2] were 0.9583 ± 0.006980, 0.9245 ± 0.01162 and 1.454 ± 1.702. The UNET_HCHD[3] model produced DSC, JI and HSD of 0.9601 ± 0.006630, 0.9274 ± 0.01132 and 1.464 ± 1.716, while 3DGC[4] produced 0.9139 ± 0.2039, 0.8809 ± 0.2152 and 1.754 ± 2.064. For DSC and JI, UNET_HCHD[3] produced the highest values, followed by and UNET_HD[2] and 3DGC[4]. The UNET_HC[1] model produced the lowest DSC and JI among the four models. For HSD, UNET_HD[2] produced the shortest surface distance between the segmented and true surface distances, followed by UNET_HCHD[3], UNET_HC[2] and 3DGC[4].

3.3. BM segmentation results of the diseased group per individual subject

When validated with the 28 diseased subjects, the means and standard deviations of DSC, JI and HSD for UNET_HC[1] were 0.8303 ± 0.1073, 0.7738 ± 0.1013 and 2.124 ± 0.5661, respectively; those for UNET_HD[2] were 0.9502 ± 0.02173, 0.9128 ± 0.03243 and 1.589 ± 0.2919. The UNET_HCHD[3] model produced DSC, JI and HSD of 0.9502 ± 0.02171, 0.9127 ± 0.03276 and 1.646 ± 0.3193, while those of 3DGC[4] were 0.8886 ± 0.2179, 0.8443 ± 0.2352, and 2.005 ± 2.375 (Table 2). For DSC and JI, UNET_HD[2] produced the highest values, followed by UNET_HCHD[3], 3DGC[4] and UNET_HC[1]. For HSD, UNET_HD[2] produced the highest values, followed by UNET_HCHD[3] and 3DGC[4]. The UNET_HC[1] model produced the longest surface distance between the segmented and true surfaces.

3.4. Evaluations of model performances and effects of data used for training and validation

Overall, the four models we evaluated produced different segmentation results. First, the U-Net model segmented lumbar BM better than the semi-automatic 3D Grow-Cut algorithm, except when the model trained with the healthy BM (UNET_HC[1]) was used to segment BM of the diseased subjects; the performance was better using 3DGC[4] (Table 3). Second, the model trained with the diseased subjects (UNET_HD[2]) produced better results than that trained with healthy subjects (UNET_HC[1]). For DSC and HSD, the segmentation results were statistically different among the four models ($p < 0.0001$) except between UNET_HC[1] and 3DGC[4]. For JI, the results were also statistically different ($p < 0.0001$) among the models except between UNET_HD[2] and UNET_HCHD[3]. Third, our models segmented healthy lumbar BM better than diseased BM, and the difference was statistically significant ($p < 0.001$) (Table 3). Fig. 4 displays the representative slices of a healthy and diseased BM segmented by all four models we evaluated.

4. Discussion

In this study, we proposed a fully automatic segmentation model for lumbar BM using sagittal, high resolution T1-weighted MRI. Our model was constructed using 2D U-Net, which was a U-shaped convolutional network architecture designed for fast and precise image segmentation. We achieved a mean DSC of 96.02% when validated with the healthy group and 95.02% when validated with the diseased group, which were higher than most of the previously reported DSCs for vertebral segmentation using CT or MRI. Specifically, the cervical vertebrae

Table 3
The bone marrow segmentation results of the four independently trained segmentation models when validated with healthy and diseased subjects. The means and standard deviations are recorded for each model. (Abbreviations: UNET_HC[1] = U-Net model trained with the healthy subjects only, UNET_HD[2] = U-Net model trained with the subjects with hematologic diseases, UNET_HCHD[3] = U-Net model trained with both healthy subjects and those with hematologic diseases, 3DGC[4] = 3-dimensional Grow-Cut segmentation algorithm, DSC = Dice similarity coefficient, JI = Jaccard Index, HSD = Hausdorff surface distance). Statistical difference ($p < 0.0001$ after Bonferroni correction) among different models are marked with asterisk*, and between validation with HC and with HD are marked with dagger†.

	Validation	Segmentation Models				Overall P-value	Pairwise Comparisons					
		UNET_HC[1]	UNET_HD[2]	UNET_HCHD[3]	3DGC[4]		1–2	1–3	1–4	2–3	2–4	3–4
DSC	HC	0.9415 ± 0.0705	0.9583 ± 0.0516	0.9602 ± 0.0486	0.9139 ± 0.2039	<0.001	*	*	0.043	*	*	*
	HD	0.8303 ± 0.107†	0.9502 ± 0.0217	0.9502 ± 0.0217†	0.8886 ± 0.2179†		*	*	*	1.00	*	*
JI	HC	0.8976 ± 0.1220	0.9245 ± 0.0919	0.9274 ± 0.0874	0.8809 ± 0.2152	<0.001	*	*	0.681	*	*	*
	HD	0.7738 ± 0.1013†	0.9128 ± 0.0324†	0.9127 ± 0.0328†	0.8443 ± 0.2352†		*	*	*	1.00	*	*
HSD (mm)	HC	1.605 ± 1.883	1.454 ± 1.702	1.464 ± 1.716	1.754 ± 2.064	0.0410	*	*	*	1.00	*	*
	HD	2.124 ± 0.566†	1.589 ± 0.2919†	1.646 ± 0.319†	2.005 ± 2.375†		*	*	*	1.00	*	*

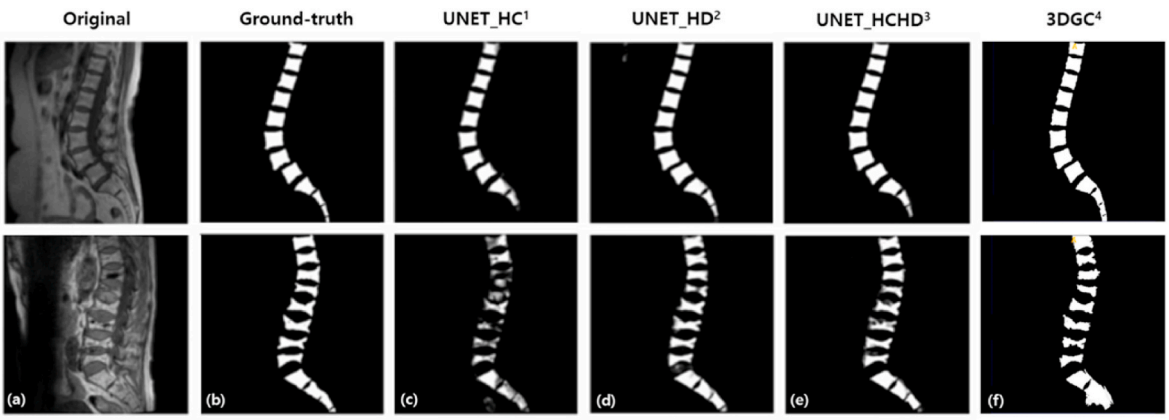


Fig. 4. Lumbar BM segmentation results by 2D U-Net trained with different data types. (a) A slice of a sagittal, high-resolution T1-weighted MRI of a healthy (top) and diseased lumbar spine (bottom), (b) lumbar BM delineated by an experienced radiologist, segmentation results by 2D U-Net models trained with (c) healthy subjects only (UNET_HC[1]), (d) diseased subjects only (UNET_HD[2]), (e) healthy and diseased subjects combined (UNET_HCHD[3]), and (f) by a semi-automatic 3-dimensional Grow-Cut algorithm (3DGC[4]).

segmentation using CT images achieved a mean DSC of 94.37% [20], and spine segmentation on CT images achieved the mean DSC of 94% [21]. The mean DSC of the web-based spine segmentation using CT images was 90.4% [22] and that of the 3D vertebral body segmentation using CT and MRI was 88.7% [25]. Although the direct comparison of different imaging modalities, segmentation methods and targets is difficult and should be interpreted with caution, our experimental results demonstrated a competitive performance on lumbar BM segmentation when compared with a number of previously-reported methods for vertebral segmentation.

Our fully-automatic model also achieved higher DSC and JI than those from the semi-automatic Grow-Cut algorithm. The U-Net model also reduced HSD, lessening inconsistency with the ground-truth. Although the Grow-Cut algorithm segmented lumbar BM with high accuracy, its advantage had been limited to healthy BM; for diseased BM with more complex shapes and wider ranges of heterogeneous signal intensity, the Grow-Cut algorithm either failed to separate BM from the disc regions or delineated BM with untidy border lines. Our U-Net model provided a more time-reducing and cost-effective way of segmenting lumbar BM with clearer margins while efficiently excluding other structures adjacent to BM.

To the best of our knowledge, this was the first study to experiment with the different data characteristics. Our segmentation results were not only affected by the methods of the models but also by the data characteristics used to train the model. The segmentation results noticeably improved when we trained U-Net with the diseased subjects, which involved lumbar BM of more complex shapes and heterogeneous signal intensity than those of the healthy subjects, which had fairly consistent rectangular shapes and uniform signal intensities across BM. The U-Net model was least effective when trained with healthy BM and was validated for diseased BM segmentations, whose results were even inferior to those of the Grow-Cut algorithm (Table 3). Our study thus demonstrated the importance of data characteristics that should be designed for model training. For lumbar BM segmentation, we insist training U-Net models with diseased cases that involved wider variety of BM shapes and signal intensity ranges was more suitable than those with the healthy subjects only.

It is to be noted that no significant effect of using both healthy and diseased BM was observed on model training. For healthy BM segmentation, UNET_HCHD[3] produced the results only slightly better than those from UNET_HD[2]. For diseased BM segmentation, UNET_HCHD[3] and UNET_HD[2] produced similar outcomes; our results infer that the number of subjects used for training did not significantly improve segmentation results and that the healthy BM might not even be sufficient for segmenting diseased lumbar BM. Our study suggests the needs for training healthy and diseased BM independently, while the number and type of subjects required for segmentation of lumbar BM with the most optimal results are subject to further studies.

5. Study limitations

There were several limitations in our study. First, we only performed 2D U-Net, which involved a slice-based learning rather than a volume-based learning. Unlike other structures in the body, sagittal BM on T1-weighted images had fairly uniform inter-slice information, which made 2D U-Net feasible for BM segmentation. Constructing 3D U-Net models for high resolution images that involves training of the entire field-of-view is resource-intensive and requires much better hardware settings. We insist that the 2D U-Net model was simple and cost-effective for segmenting lumbar BM using sagittal, high-resolution T1-weighted images. Second, our segmentation results might have excluded some marginal BM areas with partial volumes, segmenting only parts of the BM regions as opposed to the whole BM delineation. However, the marginal BM does not usually contain critical information for diagnosis of BM diseases or fracture. It would rather introduce erroneous information from bone cortex or adjacent soft tissues other than BM itself. Finally, this was a single center study, and we could not perform external validation for our model. Future studies should include dataset from multi-institutions with diverse image qualities and acquisition parameters to ensure generalizability of the model.

6. Conclusion

We constructed a fully automatic lumbar BM segmentation model for a high-resolution T1-weighted MRI using U-Net, which outperformed most of the previously reported approaches and the existing semi-automatic algorithm. It was the first study to experiment with different data characteristics. Our study suggests the need for cautious data selection before training the segmentation model, which produced different segmentation accuracies depending on the data characteristics used for model training.

Code and model availability statement

Code for the U-Net model construction and U-Net models for lumbar BM segmentation using sagittal, high-resolution T1-weighted images is provided as part of the replication package and is available at https://github.com/ohjinhwang/UNET for review.

Funding

This research was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Korea government (MSIT) (2019R1C1C1007122).

Summary

The purpose of this study was to construct a fully automatic segmentation model for lumbar bone marrow (BM) in a high resolution T1-weighted magnetic resonance imaging. Forty-four healthy controls and 56 subjects with various types of hematologic diseases received MRI of the lumbar spines, and lumbar BM of each subject was manually delineated by an experienced radiologist as a ground truth. To compare our model performances with an existing method, we segmented lumbar BM of each subject with a 3D Grow-Cut algorithm, which is a semi-automatic algorithm to segment the area of interest from multiple slices of an image. Our fully automatic segmentation models were constructed using 2D U-Net, which is a convolutional neural network used for image segmentation with a symmetrical U-shape with a left-hand contracting path to capture context of the input image followed by an expansive right-hand path to precisely localize the contextual information from the contracting path. The U-Net models were trained using lumbar BM of healthy controls (UNET_HC[1]), lumbar BM of diseased subjects (UNET_HD[2]), and of healthy and diseased subjects combined (UNET_HCHD[3]), respectively. Each U-Net model was validated using the healthy controls and diseased subjects, separately. A repeated-measures analysis of variance was performed to compare segmentation accuracies with 2 validation cohorts among U-Net trained with healthy subjects (UNET_HC[1]), U-Net trained with diseased subjects (UNET_HD[2]), U-Net trained with all subjects including both healthy and diseased subjects (UNET_HCHD[3]), and 3-dimensional Grow-Cut algorithm (3DGC[4]). When validated with the healthy subjects, UNET_HC[1], UNET_HD[2], UNET_HCHD[3] and 3DGC[4] achieved the mean and standard deviation of the Dice Similarity Coefficient (DSC) of 0.9415 ± 0.07056, 0.9583 ± 0.05146, 0.9602 ± 0.0486 and 0.9139 ± 0.2039, respectively. When validated with the diseased subjects, DSCs of UNET_HC[1], UNET_HD[2], UNET_HCHD[3] and 3DGC[4] were 0.8303 ± 0.1073, 0.9502 ± 0.0217, 0.9502 ± 0.0217 and 0.8886 ± 0.2179, respectively. Overall, U-Net segmented lumbar BM better than the existing, semi-automatic 3DGC[4] ($P < 0.0001$), and it segmented healthy lumbar BM better than the diseased BM ($P < 0.001$). Our study also showed that UNET_HD[2] segmented lumbar BM of both healthy and diseased subjects better than UNET_HC[1] ($P < 0.0001$). Our model achieved competitive results compared with a number of previously-reported methods for vertebral segmentation and the semi-automatic 3DGC[4]. This was also the first study to experiment with different data characteristics to suggest the need for cautious data selection before training segmentation models. We insist training U-Net with diseased cases that involved a wider variety of BM shapes and signal intensity ranges was more suitable than those with the healthy subjects only. In conclusion, our model provided with an accurate, fully automatic and cost-effective way of lumbar BM segmentation using a high resolution T1-weighted MRI.

Declaration of competing interest

- None of the authors or authors' institutions have financial and personal relationships with other people or organisations that could inappropriately influence (bias) their work.
- This research was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Korea government (MSIT) (2019R1C1C1007122)

References

[1] A. Birbrair, P.S. Frenette, Niche heterogeneity in the bone marrow, Ann. N. Y. Acad. Sci. 1370 (1) (2016 Apr) 82, https://doi.org/10.1111/nyas.13016.
[2] C.J. Hanrahan, L.M. Shah, MRI of spinal bone marrow: part 2, T1-weighted imaging-based differential diagnosis, AJR 197 (6) (2011 Dec) 1309–1321, https://doi.org/10.2214/AJR.11.7420.
[3] C. Zhou, H.P. Chan, Q. Dong, D.R. Couriel, A. Pawarode, L.M. Hadjiiski, J. Wei, Quantitative analysis of MR imaging to assess treatment response for patients with multiple myeloma by using dynamic intensity entropy transformation: a preliminary study, Radiology 278 (2) (2016 Feb) 449–457, https://doi.org/10.1148/radiol.2015142804.
[4] V. Chianca, R. Cuocolo, S. Gitto, D. Albano, I. Merli, J. Badalyan, M.C. Cortese, C. Messina, A. Luzzati, A. Parafioriti, F. Galbusera, Radiomic machine learning classifiers in spine bone tumors: a multi-software, multi-scanner study, Eur. J. Radiol. 137 (2021 Apr 1) 109586, https://doi.org/10.1016/j.ejrad.2021.109586.
[5] E.J. Hwang, J.Y. Jung, S.K. Lee, S.E. Lee, W.H. Jee, Machine learning for diagnosis of hematologic diseases in magnetic resonance imaging of lumbar spines, Sci. Rep. 9 (1) (2019 Apr 15) 1–9, https://doi.org/10.1038/s41598-019-42579-y.
[6] J. Liu, P. Zeng, W. Guo, C. Wang, Y. Geng, N. Lang, H. Yuan, Prediction of high-risk cytogenetic status in multiple myeloma based on magnetic resonance imaging: utility of radiomics and comparison of machine learning methods, J. Magn. Reson. Imag. 54 (4) (2021 Oct) 1303–1311, https://doi.org/10.1002/jmri.27637.
[7] E.K. Pauwels, A.W. Coumou, M. Kostkiewicz, K. Kairemo, [^{18}F]fluoro-2-deoxy-d-glucose positron emission tomography/computed tomography imaging in oncology: initial staging and evaluation of cancer therapy, Med. Princ. Pract. 22 (5) (2013) 427–437, https://doi.org/10.1159/000346303.
[8] C.L. Hoad, A.L. Martel, R. Kerslake, M. Grevitt, A 3D MRI sequence for computer assisted surgery of the lumbar spine, Phys. Med. Biol. 46 (8) (2001 Aug) N213–N220, https://doi.org/10.1088/0031-9155/46/8/403.
[9] N. Sollmann, M. Dieckmeyer, S. Schlaegar, A. Rohrmeier, J. Syvaeri, M. N. Diefenbach, D. Weidlich, S. Ruschke, E. Klupp, D. Franz, E.J. Rummeny, C. Zimmer, J.S. Kirschke, D.C. Karampinos, T. Baum, Associations between lumbar vertebral bone marrow and paraspinal muscle fat compositions—an investigation by chemical shift encoding-based water-fat MRI, Front. Endocrinol. 9 (2018 Sep 28) 563, https://doi.org/10.3389/fendo.2018.00563.
[10] E. Burian, K. Subburaj, M.R.K. Mookiah, A. Rohrmeier, D.M. Hedderich, M. Dieckmeyer, M.N. Diefenbach, S. Ruschke, E.J. Rummeny, C. Zimmer, J. S. Kirschke, D.C. Karampinos, T. Baum, Texture analysis of vertebral bone marrow using chemical shift encoding-based water-fat MRI: a feasibility study, Osteoporos. Int. 30 (6) (2019 Jun) 1265–1274, https://doi.org/10.1007/s00198-019-04924-9.
[11] E. Burian, A. Rohrmeir, S. Schlaeger, M. Dieckmeyer, M.N. Diefenbach, J. Syvari, E. Klupp, D. Weidlich, C. Zimmer, E. Rummeny, C.K. Dimitrios, J.S. Kirschke, T. Baum, Lumbar muscle and vertebral bodies segmentation of chemical shift encoding-based water-fat MRI: the reference database MyoSegmenTUM spin, BMC Muscoskel. Disord. 20 (1) (2019 Apr 09) 152, https://doi.org/10.1186/s12891-019-2528-x.
[12] S.H. Huang, Y.H. Chu, S.H. Lai, C.L. Novak, Learning-based vertebra detection and iterative normalized-cut segmentation for spinal MRI, IEEE Trans. Med. Imag. 28 (10) (2009 Oct) 1595–1605, https://doi.org/10.1109/TMI.2009.2023362.
[13] Z. Peng, J. Zhong, W. Wee, J.H. Lee, Automated vertebra detection and segmentation from the whole spine MR images, 2005, IEEE Eng. Med. Biol. Soc. (2005) 2527–2530, https://doi.org/10.1109/IEMBS.2005.1616983.
[14] J. Carballido-Gamio, S.J. Belongie, S. Majumdar, Normalized cuts in 3-D for spinal MRI segmentation, IEEE Trans. Med. Imag. 23 (1) (2004 Jan) 36–44, https://doi.org/10.1109/TMI.2003.819929.
[15] J. Egger, T. Kapur, T. Dukatz, Kolodziej, D. Zukic, B. Freisleben, C. Nimsky, Square-Cut: a segmentation algorithm on the basis of a rectangular shape, PLoS One 7 (2) (2012 Feb 21), e31064, https://doi.org/10.1371/journal.pone.0031064.
[16] C.L. Hoad, A.L. Martel, Segmentation of MR images for computer-assisted surgery of the lumbar spine, Phys. Med. Biol. 47 (19) (2002 Oct 7) 3503–3517, https://doi.org/10.1088/0031-9155/47/19/305.
[17] C. Davatzikos, D. Liu, D. Shen, E.H. Herskovits, Spatial normalization of spine MR images for statistical correlation of lesions with clinical symptoms, Radiology 224 (3) (2002 Sep) 919–926, https://doi.org/10.1148/radiol.2243011266.
[18] D. Stern, B. Likar, F. Pernuš, T. Vrtovec, Parametric modelling and segmentation of vertebral bodies in 3D CT and MR spine images, Phys. Med. Biol. 56 (23) (2011 Dec 7) 7505–7522, https://doi.org/10.1088/0031-9155/56/23/011.
[19] V.L. Tran, H. Lin, H. Liu, MBNet: a multi-task deep neural network for semantic segmentation and lumbar vertebra inspection on X-ray images, In Proceedings of the Asian Conference on Computer Vision (2020).
[20] H.J. Bae, H. Hyun, Y. Byeon, et al., Fully automated 3D segmentation and separation of multiple cervical vertebrae in CT images using a 2D convolutional neural network, Comput. Methods Progr. Biomed. 184 (2020) 105–119, https://doi.org/10.1016/j.cmpb.2019.105119.
[21] M. Vania, D. Mureja, D. Lee, Automatic spine segmentation from CT images using convolutional neural network via redundant generation of class labels, J. Comput. Des. Eng. 6 (2) (2019 Apr) 224–232.
[22] Y.J. Kim, B. Ganbold, K.G. Kim, Web-based spine segmentation using deep learning in computed tomography images, J. Healthc. Inform. Res. 26 (1) (2020 Jan) 61–67, https://doi.org/10.4258/hir.2020.26.1.61.
[23] H. Tang, X. Pei, S. Huang, X. Li, C. Liu, Automatic lumbar spinal CT image segmentation with a dual densely connected U-net, IEEE Access 8 (2020) 89228–89238.
[24] D. Gawel, P. Glowka, T. Kotwicki, M. Nowak, Automatic spine tissue segmentation from MRI data based on cascade of boosted classifiers and active appearance

model, 2018, BioMed Res. Int. (2018 Apr 29) 7952946, https://doi.org/10.1155/2018/7952946.

[25] C. Chu, D.L. Belavy, G. Armbrecht, M. Bansmann, D. Felsenberg, Fully automatic localization and segmentation of 3D vertebral bodies from CT/MR images via a learning-based method, PLoS One 10 (11) (2015 Nov 23), e1043327, https://doi.org/10.1371/journal.pone.0143327.

[26] P. Tosi, Diagnosis and treatment of bone disease in multiple myeloma: spotlight on spinal involvement, 2013, Scientifica (Cairo) (2013) 104546, https://doi.org/10.1155/2013/104546.

[27] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional Networks for Biomedical Image Segmentation in Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015, 2015.

[28] J. Egger, C. Nimsky, X. Chen, Vertebral Body Segmentation with GrowCut: Initial Experience, Workflow and Practical Application, vol. 5, SAGE Open Med, 2017 Nov 13, https://doi.org/10.1177/2050312117740984.31, 2050312117740984.

[29] G. Grinstein, C. Jayaprakash, Y. He, Statistical mechanics of probabilistic cellular automata, Phys. Rev. Lett. 55 (23) (1985 Dec 2) 2527–2530, https://doi.org/10.1103/PhysRevLett.55.2527.